

Software Design Description 200

ESP32-DevKitV1 Main Controller – Overview

Overview

The ESP32-DevKitV1 is the central hub of the trail camera system. It receives raw image frames from the ESP32-CAM over SPI, forwards them to the Tang Nano 9K FPGA over a second SPI bus for WHT compression, receives compressed coefficients back from the FPGA, packages them into LoRa packets, and transmits over UART to the RYLR998 module. It also manages peripheral sensors (temperature/humidity), a stepper motor (via FPGA I2C GPIO expander), and an interrupt-driven PIR motion detector.

Hardware: ESP32-DevKitV1 | No PSRAM | Dual SPI buses | UART to LoRa | I2C to FPGA

Module Hierarchy

main.c	Entry point, init, task creation, heap monitor
---- src/interprocessor/	
---- cam_spi.c	SPI slave - receives frames from ESP32-CAM
---- fpga_spi.c	SPI master - sends frames to FPGA, reads coefficients
---- lora_uart.c	UART interface to RYLR998 LoRa module
---- lora_bench.c	LoRa bench test (TEST_MODE_LORA_BENCH)
---- i2c_bus.c	I2C master bus driver
---- src/peripherals/	
---- aht20.c	Temperature/humidity sensor (I2C)
---- fpga_gpio.c	FPGA I2C GPIO expander client (stepper + buttons)
---- motor.c	Stepper motor control sequences
---- pir.c	PIR motion detector (interrupt-driven)
---- sensors_temp_humidity.c	Sensor task - periodic AHT20 reads
---- src/utilities/	
---- image_processor.c	Image processing pipeline task
---- isr.c	ISR semaphore initialization
---- utils.c	Common utilities

Operating Modes

Production Mode (default)

Five FreeRTOS tasks run concurrently:

Task	Priority	Stack	Core	Purpose
cam_rx	High	4 KB	Any	Receive frames from ESP32-CAM over SPI
img_proc	Medium	8 KB	Any	Process received frames, forward to FPGA
lora_rx	Medium	4 KB	Any	Receive LoRa packets (commands from S3)
sensors	Low	4 KB	Any	Periodic AHT20 temperature/humidity reads
heap_mon	(main)	—	0	Heap usage logging every 10s

LoRa Bench Mode (`#define TEST_MODE_LORA_BENCH`)

Only the `lora_bench` task runs. All camera, FPGA, and sensor initialization is skipped. Used for RF link validation with the S3 echo slave. See SDD004 for full bench test protocol.

Other Test Modes

- `TEST_MODE_FPGA_PATTERNS` — sends test patterns to FPGA over SPI
 - `TEST_MODE_FPGA_GPIO` — exercises FPGA I2C GPIO expander
 - `TEST_MODE_LORA_LOOPBACK` — local LoRa echo test
 - Mutually exclusive (enforced by `#error` in `config.h`)
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Interprocessor Communication

CAM → DevKitV1 (SPI Slave)

- DevKitV1 acts as SPI slave receiving frames from ESP32-CAM
- Frame protocol: 12-byte header (magic `0xCAFEDEED`, size, checksum) + raw pixel data
- See SDD100 for transmit-side details

DevKitV1 → FPGA (SPI Master)

- DevKitV1 acts as SPI master sending frames to Tang Nano 9K
- FPGA receives via `esp_interface.v` SPI slave
- **TODO:** Add SPI read transaction to receive compressed coefficients from FPGA (SOMI path, SDD401)

DevKitV1 → LoRa (UART)

- RYL98 module on UART2 (GPIO17 TX, GPIO16 RX) at 115200 baud
- AT command protocol: `AT+SEND=<dest>,<len>,<data>`
- Module address: 2 (sender), destination: 1 (S3 receiver)
- Network ID: 6 (must match S3)
- See SDD003 for link budget, SDD004 for AT protocol details

DevKitV1 – FPGA (I2C)

- I2C master on I2C port 0
 - Communicates with FPGA's `i2c_gpio_expander.v` at address `0x27`
 - Writes: stepper motor coil patterns (GPIO out bits [3:0])
 - Reads: joystick button state (GPIO in bits [2:0])
 - External 4.7k pull-ups on SDA/SCL
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Peripherals

AHT20 Temperature/Humidity Sensor

- I2C address: `0x38`
- Polled periodically by `sensors_task`
- Data logged, available for environmental metadata

PIR Motion Detector

- Interrupt-driven on GPIO 25
- ISR signals a semaphore, processed by main pipeline
- Triggers camera capture sequence

Stepper Motor

- 4-wire unipolar stepper
- Driven via FPGA I2C GPIO expander (coils on gpio_out[3:0])
- Pan/tilt positioning for multi-position reference frame system

Configuration Reference

Key defines from `config.h`:

Macro	Value	Notes
IMAGE_MAX_SIZE_BYTES	1 MB	Max image buffer
IMAGE_QUEUE_LENGTH	3	FreeRTOS queue depth
SPI_BUFFER_SIZE	4096	SPI transfer chunk
UART_BUFFER_SIZE	2048	LoRa UART buffer
LORA_NETWORK_ID	6	Must match S3
LORA_ADDRESS_SENDER	2	DevKitV1 address
BENCH_DEST_ADDR	1	S3 address
FPGA_I2C_PORT	0	I2C bus number
FPGA_I2C_CLK_HZ	400000	I2C fast mode

Source Files

File	Purpose	Status
src/main.c	Init, task creation, heap monitor	Done
src/interprocessor/cam_spi.c	SPI slave RX from ESP32-CAM	Done
src/interprocessor/fpga_spi.c	SPI master to FPGA	Done (TX only, RX TODO)
src/interprocessor/lora_uart.c	LoRa UART AT command interface	Done
src/interprocessor/lora_bench.c	LoRa bench test protocol	Done
src/interprocessor/i2c_bus.c	I2C master driver	Done
src/peripherals/aht20.c	AHT20 temp/humidity sensor	Done
src/peripherals/fpga_gpio.c	FPGA I2C GPIO expander client	Done
src/peripherals/motor.c	Stepper motor sequences	Done
src/peripherals/pir.c	PIR motion detector ISR	Done
src/peripherals/sensors_temp.c	Temperature sensing task	Done
src/utilities/image_processing.c	Image processing pipeline	Done
src/utilities/isr.c	ISR semaphore init	Done
src/utilities/utils.c	Common utilities	Done

References

- SDD000 — Development Standards
- SDD003 — LoRa Link Budget
- SDD004 — LoRa Bench Test & Debug
- SDD100 — ESP32-CAM
- SDD400 — FPGA Subsystem
- SDD401 — FPGA WHT Pipeline